

S-BOX AND IMAGE ENCRYPTION ANALYSIS REPORT

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Analysis Date: 2025-10-17_20-20-35
S-box Name: DEFAULT 16x16 S-box
S-box Size: 8x8 (256 elements)
Test images: Astronaut, Cameraman, Coins

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PART I: S-BOX CRYPTANALYSIS
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1. BASIC S-BOX INFORMATION
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S-box Name: DEFAULT 16x16 S-box
Size: 8x8
Number of elements: 256
Bijective S-box: Yes

2. CRYPTOGRAPHIC PROPERTIES
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2.1. NONLINEARITY
    • Minimum nonlinearity: 106
    • Maximum nonlinearity: 108
    • Average nonlinearity: 107.2500
    • VBF nonlinearity: 94

2.2. DIFFERENTIAL PROPERTIES
    • Differential uniformity: 10

2.3. AVALANCHE PROPERTIES (SAC)
    • SAC average: 0.499268
    • SAC deviation: 0.034424

2.4. LINEAR PROPERTIES
    • Maximum linear approximation probability (LAP): 0.13281250
    • Average linear approximation probability: 0.02468791
    • Maximum linear bias: 34
    • Average linear bias: 6.3201
    • Maximum autocorrelation: 64

2.5. BIT INDEPENDENCE (BIC-NL)
    • Minimum BIC-NL: 98
    • Maximum BIC-NL: 108
    • Average BIC-NL: 103.9286

2.6. ALGEBRAIC PROPERTIES
    • Algebraic degree: 8

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PART II: IMAGE ENCRYPTION ANALYSIS
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3. SUMMARY RESULTS
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Average PSNR (Encryption)      : 7.31 dB
Average NPCR                   : 99.62%
Average UACI                   : 35.87%
Average MSE                    : 12279
Average MAE                    : 91.46
Average Correlation Reduction: 0.7899

4. DETAILED IMAGE ANALYSIS
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--- Astronaut Image ---

Performance Metrics:
  PSNR (Encryption): 6.34 dB
  PSNR (Decryption): inf dB
  MSE               : 15096
  MAE               : 102.53

Security Metrics:
  NPCR              : 99.45%
  UACI              : 40.21%

Correlation Analysis:
  Horizontal: 0.9840 → 0.2057
  Vertical   : 0.9862 → 0.2190
  Diagonal   : 0.9758 → 0.1935

Attack Resistance:
  Occlusion Attack PSNR: 11.79 dB
  Noise Attack PSNR    : 7.37 dB

--- Cameraman Image ---

Performance Metrics:
  PSNR (Encryption): 7.37 dB
  PSNR (Decryption): inf dB
  MSE              : 11909
  MAE              : 90.61

Security Metrics:
  NPCR             : 99.77%
  UACI             : 35.53%

Correlation Analysis:
  Horizontal: 0.9781 → 0.2511
  Vertical   : 0.9853 → 0.2482
  Diagonal   : 0.9712 → 0.2281

Attack Resistance:
  Occlusion Attack PSNR: 9.79 dB
  Noise Attack PSNR    : 7.88 dB

--- Coins Image ---

Performance Metrics:
  PSNR (Encryption): 8.20 dB
  PSNR (Decryption): inf dB
  MSE              : 9834
  MAE              : 81.23

Security Metrics:
  NPCR             : 99.62%
  UACI             : 31.86%

Correlation Analysis:
  Horizontal: 0.9372 → 0.0729
  Vertical   : 0.9405 → 0.0585
  Diagonal   : 0.9054 → 0.0439

Attack Resistance:
  Occlusion Attack PSNR: 11.93 dB
  Noise Attack PSNR    : 8.48 dB

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PART III: EVALUATION
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5. S-BOX AND IMAGE ENCRYPTION COMPATIBILITY
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5.1. S-BOX CRYPTOGRAPHIC QUALITY
✓ High Nonlinearity quality
△ Medium Differential resistance
✓ Ideal SAC property
✓ Complete Bijective property

5.2. IMAGE ENCRYPTION EFFECTIVENESS
✓ Excellent Encryption effectiveness (PSNR: 7.31 dB)
✓ Excellent Pixel change ratio (NPCR: 99.62%)
✓ Ideal Intensity change uniformity (UACI: 35.87%)
✓ Good Correlation weakening

6. GENERAL RECOMMENDATIONS
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• Differential uniformity is too high, S-box should be optimized

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TECHNICAL INFORMATION
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S-box Type      : DEFAULT 16x16 S-box
Test Images     : Astronaut, Cameraman, Coins
Analysis Tools  : Python, NumPy, scikit-image, matplotlib
Cryptographic Metrics: Nonlinearity, DU, SAC, LAP, BIC-NL, Algebraic Degree
Image Metrics   : PSNR, MSE, MAE, NPCR, UACI, Correlation Analysis
Visualizations  : 4 PNG files created
Attack Simulation : Occlusion (20%) and Gaussian Noise (10%)

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CONCLUSION
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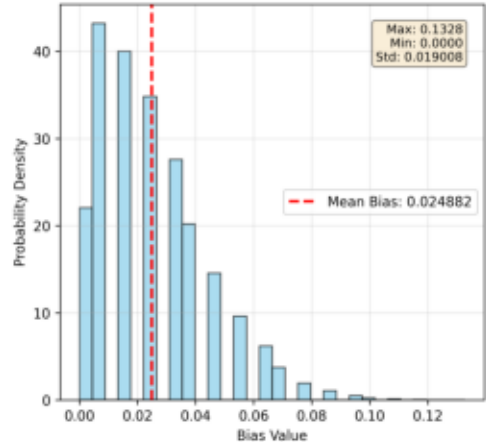
This S-box has been tested for both cryptographic properties and image encryption performance. For detailed visual analysis, see the generated PNG files.

Analysis completed: 2025-10-17_20-20-35
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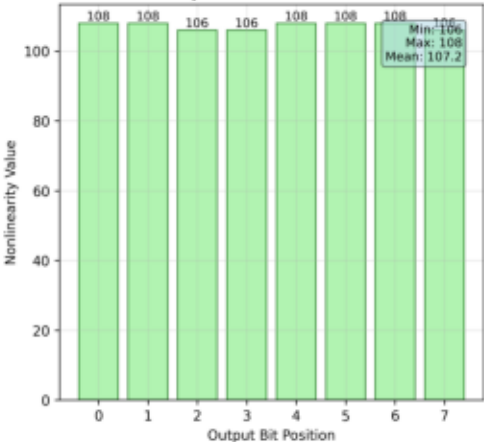
S-BOX CRYPTOGRAPHIC PROPERTIES ANALYSIS

DEFAULT 16x16 S-box - Cryptographic Properties Analysis

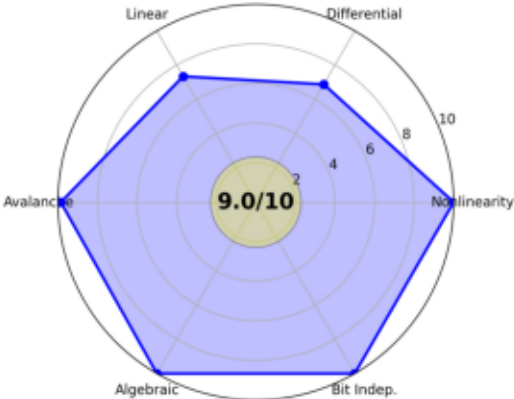
Linear Bias Distribution



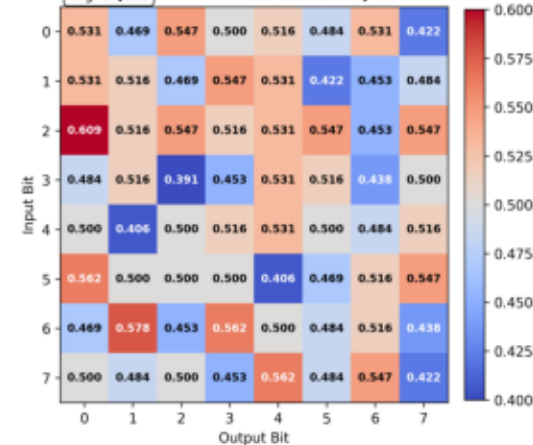
Component Nonlinearities



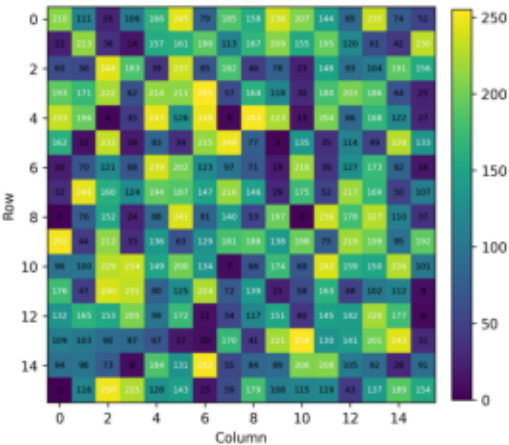
Cryptographic Strength Profile



SAC (Strict Avalanche Criterion) Matrix



S-box Value Matrix



90.1%

Cryptographic Strength
Good

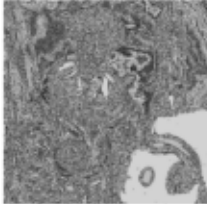
IMAGE PROCESS VISUALIZATION

IMAGE ENCRYPTION PROCESS AND HISTOGRAM ANALYSIS

Astronaut - Original



Astronaut - Encrypted



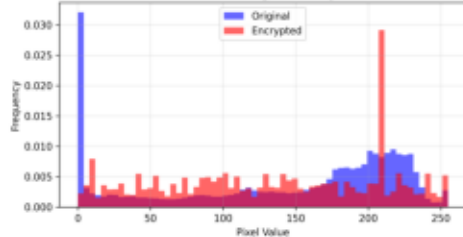
Astronaut - Decrypted
PSNR: inf dB



Astronaut - Occlusion Recovery
PSNR: 11.79 dB



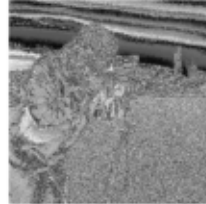
Astronaut - Histogram Comparison



Camerman - Original



Camerman - Encrypted



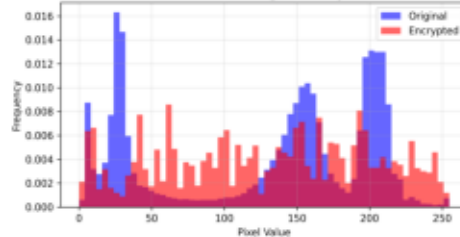
Camerman - Decrypted
PSNR: inf dB



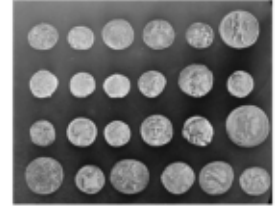
Camerman - Occlusion Recovery
PSNR: 9.79 dB



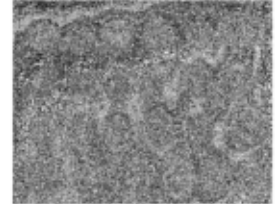
Camerman - Histogram Comparison



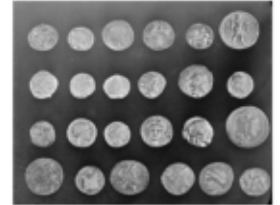
Coins - Original



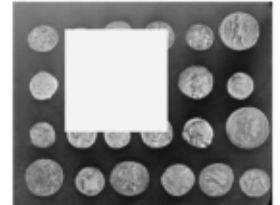
Coins - Encrypted



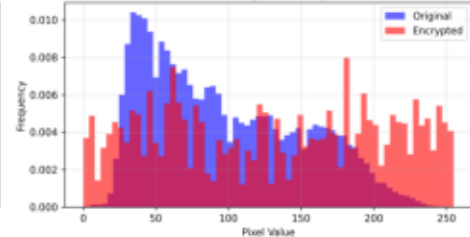
Coins - Decrypted
PSNR: inf dB



Coins - Occlusion Recovery
PSNR: 11.93 dB

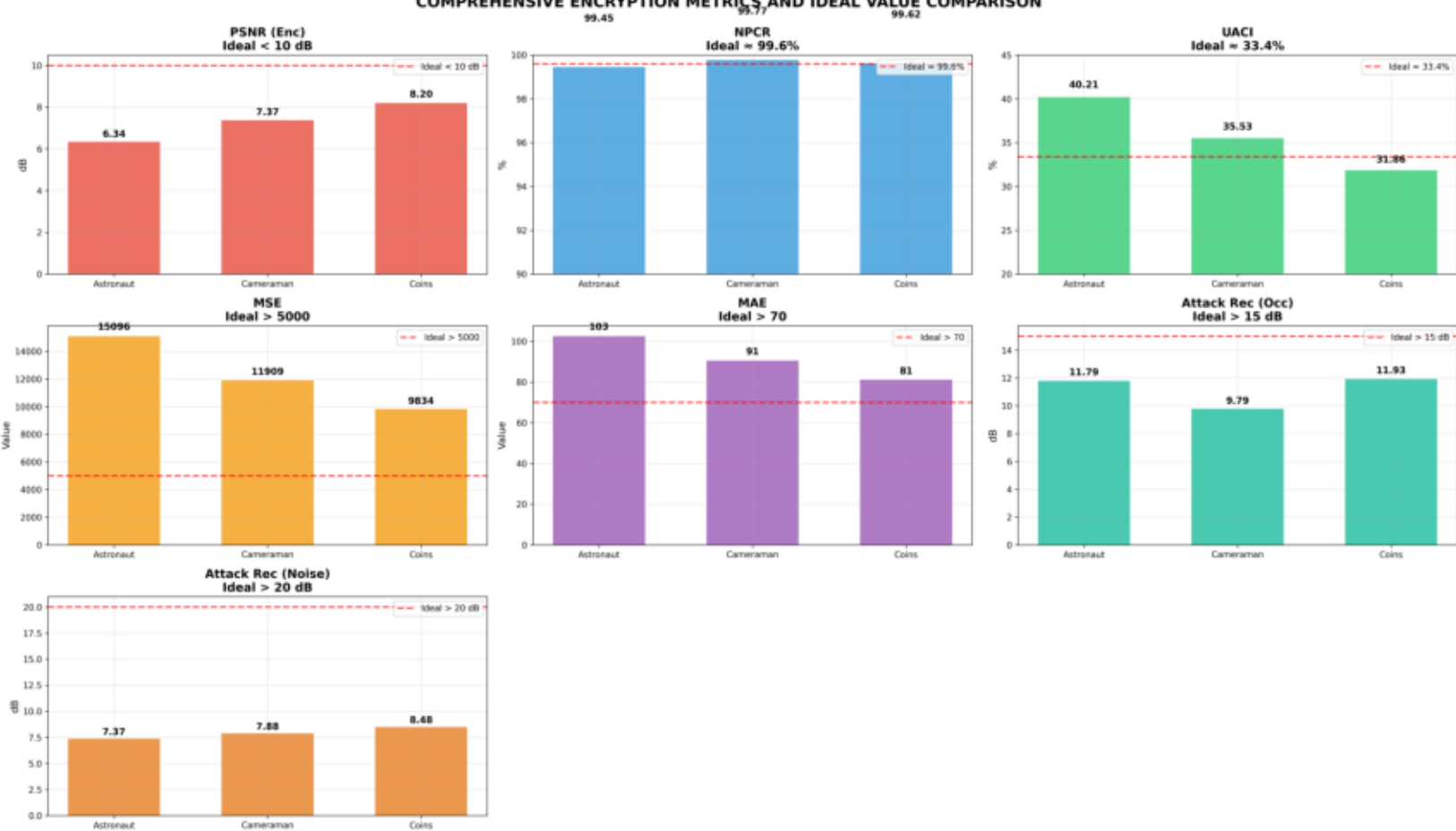


Coins - Histogram Comparison



METRICS AND IDEAL VALUE COMPARISON

COMPREHENSIVE ENCRYPTION METRICS AND IDEAL VALUE COMPARISON



CORRELATION ANALYSIS

CORRELATION ANALYSIS - ORIGINAL vs ENCRYPTED

